Week 4 Day 3 — Feature Engineering & Model Comparison

Adult Census Income dataset | New features, cross-validation and model comparison | Aqsa Saqib

1. What I did

I made 8 new features and added them to my pipeline. Then I compared three models with 5-fold cross-validation on the training data (26,048 rows). The hold-out test set was not used today.

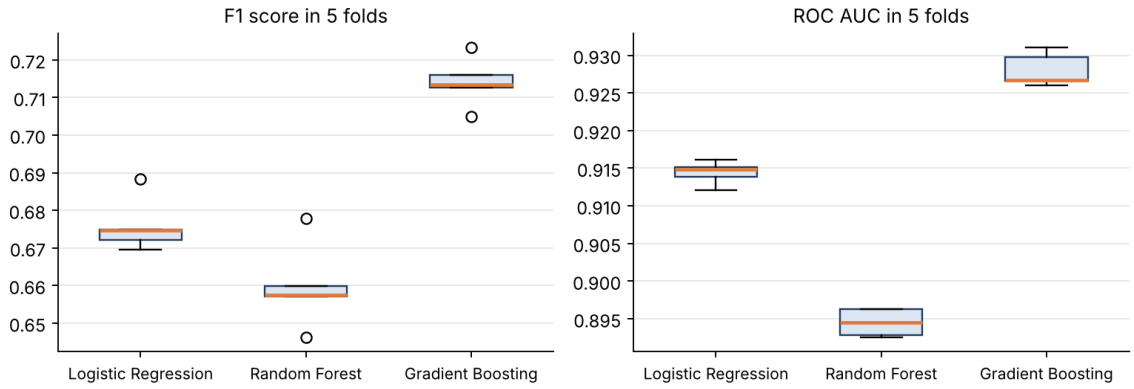
2. New features

Feature	How it is made	Why it may help	Score
is_married	marital status is married	Married people earn >50K much more often	0.107
edu_x_hours	education.num x hours.per.week	Educated people who work long hours should earn the most	0.082
log_capital_gain	log(capital.gain + 1)	Makes the few huge values smaller	0.081
age_group	age cut into 6 groups	Income goes up with age and then goes down again	0.062
higher_education	education.num >= 13	Bachelors or more was a strong signal on Day 1	0.048
hours_group	hours.per.week cut into 4 groups	Part-time, full-time and overtime workers earn differently	0.038
log_capital_loss	log(capital.loss + 1)	Same idea as capital gain	0.035
has_capital_gain	capital.gain > 0	People with any capital gain are more often high earners	0.030

- **Score** = mutual information on the training data. Higher means the feature tells more about the target.
- **Best feature:** is_married. Only 6% of unmarried people earn >50K, but 45% of married people do.
- **No data leakage:** every feature uses only the person's own row, and none of them uses the target.

3. Cross-validation results (5 folds)

Model	F1 (mean +/- std)	ROC AUC (mean +/- std)
Logistic Regression	0.676 +/- 0.007	0.914 +/- 0.002
Random Forest	0.660 +/- 0.011	0.895 +/- 0.002
Gradient Boosting	**0.714 +/- 0.007**	**0.928 +/- 0.002**



- **Gradient Boosting is the best model.** Logistic Regression is second and Random Forest is third.
- The std is very small, so the scores are stable in all 5 folds.

4. Did the new features help?

Model	F1 with old features	F1 with new features
Logistic Regression	0.658	0.676
Gradient Boosting	0.714	0.714

- The new features help Logistic Regression.
- They do not change Gradient Boosting. A tree model can make groups like "age 36-45" by itself.

5. Statistical test (top 2 models)

- **Test:** paired t-test on the 5 F1 scores of Gradient Boosting and Logistic Regression.
- **Result:** difference in mean F1 = 0.038, p-value = 0.0005.
- **Meaning:** the p-value is far below 0.05, so the difference is real and not luck. Gradient Boosting is better in all 5 folds, and a gap of about 0.04 in F1 is big enough to matter.
- 5 folds is a small sample, so this test is only a rough check.

6. Which new features mattered most?

- **Random Forest (importance):** edu_x_hours (0.092), log_capital_gain (0.081), is_married (0.054).
- **Logistic Regression (coefficients):** log_capital_gain and is_married push towards >50K. The young (17-25), old (66+) and part-time groups push towards <=50K.
- has_capital_gain and log_capital_gain carry almost the same information, so they work as a pair.
- This makes sense. Capital gain, marriage, education and working hours were also strong on Day 1 and Day 2.

7. Feature selection

Setup (Gradient Boosting)	Columns	F1	Training time per fold
All features	104	0.714	about 2 seconds
Best 30 by mutual information	30	0.692	about 12 seconds

- With only 30 columns the F1 score goes down.
- Training gets slower, not faster, because calculating mutual information takes time.

8. Recommended for tuning

Models: Gradient Boosting (best, F1 0.714) and Logistic Regression (second, simple and easy to explain).
Features: keep all the features. Removing features lowers the score and saves no time.